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Effects of functional groups and species richness on biomass constituents relevant for combustion: results from a grassland diversity experiment

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Abstract

Grassland biomass has been identified as a potential energy source. The combustion of mature and fibrous biomass, as occurs in extensive grasslands managed with low cutting frequencies, is one possible conversion technique. This study tested the relationship between plant diversity and biomass constituents relevant for combustion, as they determine energy content, energy yield and emission and corrosion risks. The biomass from a biodiversity experiment, with a species richness (SR) gradient of 1–60 species from Central European mesophilic grasslands divided into four functional groups (grasses, legumes, small and tall herbs), was harvested twice a year (in 2008 and 2009). The higher heating value (HHV) was estimated from carbon, hydrogen and oxygen contents to give insight into the energy potential of the species mixtures. The potential risk of emission and corrosion was assessed by analysing ash content, potassium, calcium, magnesium, nitrogen, sulphur and chloride content. HHV was independent of SR, and the overall mean was 18·13 MJ kg^{-1} DM. Biomass and gross energy (GE) yield were positively affected by SR. The presence of legumes in a mixture resulted in increased HHV, biomass yield and GE, irrespective of the level of SR. Annual GE varied between 59 (average of monocultures) and 152 GJ ha^{-1} $year^{-1}$ (mixture of sixty species). The concentration of ash-forming elements was generally high, suggesting a pre-treatment of the biomass prior to combustion. Emission- and corrosion-related constituents were clearly affected by the different functional groups, and sulphur and nitrogen both declined with SR. The results of this study show that high SR in experimental grassland communities is beneficial for the energy output and that legumes play a key role for the energy potential. However, identifying a functional group as being solely beneficial or disadvantageous for fuel quality was difficult.

Keywords: bioenergy, combustion, diversity, extensive grassland, functional groups, Jena

Introduction

The need to explore renewable energy sources has brought attention to different kinds of biomass (from agriculture, forestry and waste). In the European Union (twenty-seven member states), agricultural biomass is expected to have the largest energy potential in the long term (European Environment Agency, 2006). However, with increasing cultivation of energy crops, such as maize (*Zea mais* L., grown for anaerobic fermentation) and miscanthus (*Miscanthus* × *giganteus*, grown for combustion), a discussion about the competition with food production and their negative impacts on the environment (e.g. loss of biodiversity, greenhouse gas emissions owing to land-use change) has arisen (Fargione *et al.*, 2008; Petersen, 2008; Tilman *et al.*, 2009).

In the context of grassland utilization, competition with food production and environmental impacts can be minimized if biomass is taken from areas where agricultural management is receding and from land that would otherwise be abandoned. For some regions in central Europe, it has been estimated that up to one-quarter of permanent grassland will be abandoned in the near future because of structural changes in agricultural practices (Rösch *et al.*, 2007), which will mainly affect extensive grasslands on less-favourable soils. Hence, the utilization for energy production may

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provide an opportunity for regular management and maintenance of such grasslands. It could provide a way of protecting landscapes that are valuable to society because of their aesthetic and recreational functions and help to maintain their ecosystem services; i.e., nature and species conservation, groundwater collection, prevention of erosion and soil degradation (Wilkins and Vidrih, 2000), especially when species diversity is high (Minns *et al.*, 2001).

The preservation of extensive grassland usually requires 1–2 cuts per year, resulting in a fibre-rich substrate that will result in relatively low methane yields during anaerobic fermentation (Lemmer and Oechsner, 2001; Prochnow *et al.*, 2005), but still contains energy that can be accessed by other conversion techniques. Combustion has been considered as a promising technique to convert fibre-rich, extensive grassland biomass into energy (Prochnow *et al.*, 2009; Rösch *et al.*, 2009).

If combustion is chosen as a conversion technique, fuel quality becomes of great importance. It is a major determinant of the lifespan of the combustion plant and of the choice of type of plant (regarding combustion chamber layout, equipment with filter and condensation units, grate methods, etc.). While the carbon (C), hydrogen (H) and oxygen (O) content in wood from spruce (*Picea*) accounts for about 99% of the total biomass, it comprises up to 95% of the biomass from miscanthus and only 90–93% of grassland biomass, depending on the species or species composition (Hartmann, 2009). The remaining fraction consists of elements that contribute to ash formation and/or become volatile and cause emissions. The major ash-forming elements in biomass are silicon (Si), calcium (Ca), magnesium (Mg), potassium (K), sodium (Na) and phosphorus (P) (van Loo and Koppejan, 2008). These ash-forming elements, when exposed to high temperatures, either contribute to the melting and deposition of ash on the surface of the combustion chamber (slagging) or contribute to vaporization and subsequently condense in the cooler regions (fouling). However, some elements play multiple roles in these processes, which cannot easily be disentangled.

According to Hartmann (2009), the average ash-softening temperature (AST; the temperature at which a cube of pressed ash shows the first signs of softening) of wood (1426°C for spruce) is relatively high, whereas it is much lower for hay from landscape management (1061°C) and it declines further for wheat straw (998°C) and miscanthus (973°C).

Among others, the elements creating harmful emission and corrosion problems are nitrogen (N), chlorine (Cl) and sulphur (S). N concentration in the solid fuel shows a logarithmic correlation with the NO_x emissions (van Loo and Koppejan, 2008) which have been identified as one of the major environmental impacts of solid biomass combustion (Nussbaumer, 2003). Cl is responsible for the formation of HCl and, together with alkali metals, for the formation of chlorides such as KCl or NaCl. Furthermore, high Cl concentrations are expected to contribute to the formation of polychlorinated dibenzodioxines and dibenzofuranes (PCDD/F) and their concentrations were found to be higher during the combustion of herbaceous biomass than during wood combustion (Launhardt and Thoma, 2000). The S contained in the biomass will mainly transform into SO_x. While 40–90% of the S concentration is bound in the ash, the rest is emitted as SO_2 and to a minor extent as SO_3 (van Loo and Koppejan, 2008). As part of the SO_x emission process, some of the SO_2/SO_3 will react with deposited alkali chlorides and thus replace, and consequently release, Cl as gaseous Cl_2. This leads to the formation of alkali sulphates such as K_2SO_4 or $NaSO_4$, while the Cl can repeatedly react with metal parts of the plant forming, for example, $FeCl_2$, which is the source of severe corrosion (Riedl *et al.*, 1999).

In extensive grassland, where a multitude of species occurs, the fuel quality can vary considerably depending on the individual species composition. While perennial energy grasses such as switchgrass (*Panicum virgatum*), miscanthus, or reed canary grass (*Phalaris arundinacea*) have been the focus of research on solid biofuels from grassland biomass (Christian *et al.*, 2002; Lewandowski *et al.*, 2003), there is a lack in knowledge on extensive grasslands and their characteristics regarding biofuel production (Prochnow *et al.*, 2009). The energy content and fuel quality of grassland biomass have, so far, been studied over a range of different locations (Florine *et al.*, 2006; El-Nashaar *et al.*, 2009; Wachendorf *et al.*, 2009; Tonn *et al.*, 2010) creating insight into the effects of species composition and abiotic location properties on energy content and fuel quality. However, none of the above-mentioned studies had clear diversity gradients that could have outlined their results specifically as diversity effects. Florine *et al.* (2006) mentioned that the variance in chemical composition was because of the diversity of herbaceous plant species, and Tonn *et al.* (2010) subsequently found the difference between grasses and herbaceous species to be mainly responsible for the variance. In a study on forage quality, White *et al.* (2004) found fibre content to be increasing with increasing species richness (SR) in a late cut (while no trend was found in the early cut) when multiple grasslands with different SR were compared. This could be a weak indication that, along a species-richness gradient, differences in higher heating value (HHV) might be found, as HHV is mainly affected by carbon and thus by fibre content. Furthermore, they

found that crude protein content (N content × 6·25) decreased with increasing SR in both cuts, which would lead to less NO_x emission because of less N content in the fuel. However, with their experimental approach, it cannot be addressed whether these effects are related to SR *per se* or an interaction with biotic and abiotic factors of the different sites.

Therefore, the hypothesis of this study is that changes along a species-richness and functional-group gradient regarding fuel quality are to be expected and are best investigated through an experimental set-up with artificially designed diversity gradients that reduce the variance in site-specific biotic and abiotic factors.

Thus, the intention of this study was to investigate (i) how SR may affect energy content, gross energy (GE) yield and fuel quality, (ii) how energy content and fuel quality differ among the functional groups and how they affect those parameters in mixtures, and (iii) to draw conclusions from the results regarding possible emission and corrosion risks and the potential energy output that can be expected.

This was approached by choosing an experimental layout that represents extensive grassland communities of the Arrhenatherion community through artificially assembled species mixtures. The biomass investigated in this study came from a large-scale biodiversity experiment, which was also established to address the criticism provoked by previous diversity studies (Roscher *et al.*, 2004). Therefore, the choice of functional groups was approached very carefully by giving special attention to legumes and regarding them as a separate functional group, as they have been found to have disproportionate effects on ecosystem processes (Spehn *et al.*, 2002). Whereas in previous biodiversity experiments, usually three functional groups were included (grasses, herbs and legumes), it was decided in this experiment to further divide herbs into small and tall herbs. At the same time, the testing of species-richness effects was given special attention in the design of the experiment, allowing us to disentangle the effects of SR, functional-group richness (FGR) and the presence of individual functional groups as much as possible.

Materials and methods

Experimental design

This study was conducted on the Jena Experiment field site, located in the floodplain of the River Saale at the northern edge of Jena (Thuringia, Germany, 50°55N, 11°35E; 130 m a.s.l.). Mean annual air temperature in the Jena area is 9·3°C with an annual precipitation of 587 mm (Kluge *et al.*, 2000). The site had originally been grassland and was converted into arable land around 1960. Soil conditions resemble a Eutric Fluvisol (FAO, 1994), and soil texture changes from silty clay to sandy loam with increasing distance from the river.

In May 2002, a system representing an extensive Central European mesophilic grassland (Arrhenatherion community; Ellenberg, 1996) was established at the site. Sixty species typical for this type of grassland were used to create a gradient in plant SR (1, 2, 4, 8, 16 and 60) and in FGR (1, 2, 3 and 4). Functional groups were defined according to the morphological, phenological and physiological traits of the plant species as grasses (n = 16), small herbs (n = 12), tall herbs (n = 20) and legumes (n = 12) (for detailed list of plant species, see Roscher *et al.*, 2004). Eighty-two plots (20 × 20 m) were established in four blocks, the blocks accounting for the differences in soil texture. Sixteen possible combinations of SR and FGR were realized and replicated over the four blocks (Figure 1). The location of the mixtures within each block was fully randomized. The management of the site was typical of that for extensive grassland under the actual site conditions, with two cuts per year (late May and late August) and no application of fertilizer. Plots were weeded twice a year to maintain the original species composition. The experimental set-up is described in full detail by Roscher *et al.* (2004).

Biomass yield and ash

Aboveground biomass was harvested twice in the years 2008 and 2009, immediately before the first and the second cut. Three randomly placed samples of

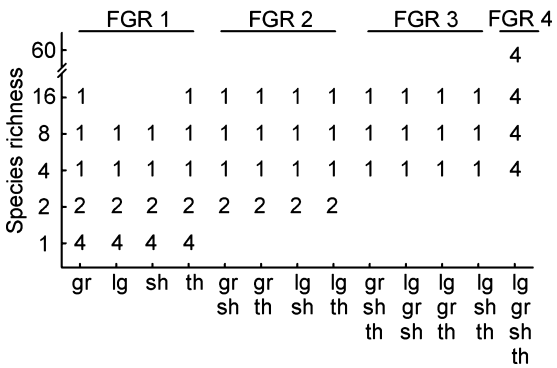

Figure 1 Combinations of functional groups and their replications (numbers) according to species-richness level. Each replication represents one of the plots used in this study; the total sum therefore is 82. The *x*-axis gives the different functional-group compositions (gr = grasses, lg = legumes, th = tall herbs, sh = small herbs). Functional-group richness (FGR) is shown on top.

20 × 50 cm were harvested 3 cm above soil surface. Biomass samples were separated into target species, dead plant material and weeds, dried (70°C, 48 h) and weighed. Mean total biomass in tonnes of dry matter (DM) per hectare (t DM ha^{-1}) was derived from the three samples. Annual biomass yield was calculated as the sum of biomass from first and second cut. Aboveground biomass was also analysed for crude ash according to standard methods (Naumann and Bassler, 2004).

Evenness

The relative abundance p_i of a species in a mixture, measured as its relative contribution to the biomass, was used to calculate the Shannon–Wiener index: $H' = \sum p_i * \ln(p_i)$ and put in relation to the total number of species S as the Shannon evenness $E = H'/\ln(S)$. Thus, E can have values between zero and one with $E = 0$ being a mixture entirely dominated by one species and with $E = 1$ being a mixture with equal proportions for all occurring species. E was not calculated for monoculture plots; thus, the number of plots with values of E was $n = 66$ in the first cut and $n = 63$ in the second (owing to three plots with SR = 2 that only had one target species present).

Energy content, ash and lignin

From the aboveground biomass, 300 g was harvested 3 cm above soil surface within the core area (3 × 3 m) of each plot, dried (70°C, 48 h) and weighed. Carbon (C), hydrogen (H) and nitrogen (N) concentrations (g kg^{-1} DM) of the biomass were analysed using an elemental analyser (vario MAX CHN, Elementar Analysensysteme GmbH, Hanau, Germany). The energy content or HHV is the amount of heat released during combustion and takes into account the latent heat of vaporization of water in the combustion product. HHV was calculated based on the concentrations of C, H and N with the empiric equation for biofuels from Friedl *et al.* (2005):

$$\mathrm{HHV} = 0 \cdot 0355 \times \mathrm{C}^2 - 23 \cdot 2 \times \mathrm{C} - 223 \times \mathrm{H} + 0 \cdot 512 \times \mathrm{C} \times \mathrm{H} + 13 \cdot 1 \times \mathrm{N} + 20\,600\,(\mathrm{kJ\,kg^{-1}DM})$$

Gross energy yield was calculated by multiplying HHV with the biomass yield.

Acid detergent lignin (ADL) was analysed, according to standard methods (Naumann and Bassler, 2004), as a chemical component rich in C and thereby potentially affecting HHV. For ADL, 100 of the 164 samples were analysed in the laboratory and then used for near-infrared calibration. With a near-infrared spectroscope (XDS Rapid Content Analyser, FOSS NIRSystems Inc., Laurel, NC, USA), the 164 samples were predicted after cross-validation (standard error of cross-validation, SECV = 1·32 g kg^{-1} DM; $R^2 = 0{\cdot}74$; residual predictive deviation, RPD = 2·35).

Constituents relevant for corrosion and emissions

From the harvested aboveground biomass (see description above), S, Cl, K, Ca and Mg (g kg^{-1} DM) were analysed by X-ray fluorescence analysis. To estimate the potential concentration of K, Ca and Mg in the ash, their DM concentration was put in relation to their ash concentration of the biomass sample. Thereby, it is assumed that the elements will all go entirely into the ash even though, in the case of K, this is highly dependent on the temperature (Knudsen *et al.*, 2004).

Statistical analysis

The Jena experiment was designed to vary SR, FGR and functional-group composition (FGC) as orthogonally as possible. However, a fully balanced design is not possible, as for example, the lowest SR cannot be combined with highest FGR. This is not an unusual situation in biodiversity experimentation and is best solved by analysing the dependent variable in an analysis of variance (ANOVA) with sequential sum of squares (Schmid *et al.*, 2007). In this type of analysis, variables that are fitted before others take up all the variation they can explain, ignoring the possibility that the later variables might also explain some of this variation (McCullagh and Nelder, 1999; Hector *et al.*, 2010). These characteristics of the analysis can be used to identify effects that are independent of what was fitted before.

To account for the gradient in soil conditions, block-wise weeding, mowing and sampling, block effects were fitted first. It can therefore be assumed that all variances that are explained by variables fitted after the block effect are independent of these.

As the main focus of this study is on the effects of increasing SR, this variable was fitted first after block. To test the effects of SR, the log-linear contrast of 1–16 species was used. The sixty species plots were used as a point of reference for highest possible diversity, but were not included in the statistical analysis. Two plots had to be omitted from the data set (one small herb monoculture plot and one tall herb monoculture plot) owing to difficulties with their establishment. Thus, the total number of plots in the statistical analysis was $n = 76$. All values used in the ANOVA were mean values of both years.

To test for the effects of FGR and FGC, these variables were fitted after SR in a first model (Tables 2A and 3A). The first model was performed as a repeated measure ANOVA to test for the effects of the two harvest dates. In the following two models, one for each cut (Tables 2 and 3B first cut and C second cut), the presence/absence of functional groups were fitted after SR to test for their individual effects.

Multiple regression analysis was conducted (on all plots including SR = 60) to estimate the influence of legume abundance and other possible functional-group abundances on GE by selecting the terms for inclusion in the model depending on standard statistical model selection methods (Draper and Smith, 1998). This implies that effect terms with $P < 0{\cdot}05$ were included according to the rules of hierarchy and marginality (Nelder, 1994; Nelder and Lane, 1995). The marginality principle (Nelder and Lane, 1995) implies that if a term appears as part of a more complex element in the model, then, in general, the term itself is not tested for significance, because the meaning of such terms is open to misinterpretation (Connolly and Wachendorf, 2001).

The initial model contained all functional-group abundance terms (lg, gr, sh, th), including their quadratic terms and all possible pairwise interactions. Functional-group abundances were tested for co-linearity (with cor (x, y, method = Pearson) in R) prior to the analysis. The possible correlations between functional-group abundances were all within $-0{\cdot}21 > r < -0{\cdot}44$ and, on average, at $r = -0{\cdot}33$ in both cuts. All statistical analyses were performed in R 2.12.2 (R Development Core Team, 2011).

Results

Energy content and relevant chemical constituents for its estimation

Among the 80 species mixtures, representing 13 different FGC, HHV varied between 16·3 and 19·2 MJ kg^{-1} DM with an overall mean of 18·13 MJ kg^{-1} DM. Among the functional-group monocultures (plots with FGR = 1), legumes had the highest HHV in both cuts, whereas grasses, tall herbs and small herbs were lower (Table 1). Diversity parameters (SR, FGR and FGC) and the presence of individual functional groups had the following effects: HHV was not affected by SR (Figure 2) or FGR, while FGC showed a substantial effect on HHV in the repeated-measures model (Table 2A). As the cut affected HHV (Table 2A), with mean values being higher in the first than in the second cut (Table 1), the two cuts were analysed separately. Here, FGC had no effect, indicating that most information of FGCs is contained in the presence/absence of the functional groups. While SR had no effects on HHV in the separate analysis of the two cuts, the presence of legumes increased HHV (Table 2,

Table 1 Energy content (HHV), carbon, hydrogen, nitrogen, lignin and ash content as means of the functional-group (FG) monocultures of both years. Values are shown separately for both cuts, and variance is indicated by standard deviation.

	Grasses	Legumes	Small herbs	Tall herbs	All FG mixtures
Parameter	**Mean ± s.d.**	**Mean ± s.d.**	**Mean ± s.d.**	**Mean ± s.d.**	**Mean ± s.d.**
HHV (MJ kg^{-1} DM)					
1st cut	18·09 ± 0·16	18·64 ± 0·21	18·29 ± 0·16	18·23 ± 0·13	18·24 ± 0·24
2nd cut	17·92 ± 0·20	18·51 ± 0·15	17·74 ± 0·17	17·79 ± 0·34	17·99 ± 0·36
Carbon (g kg^{-1} DM)					
1st cut	455·0 ± 3·5	463·6 ± 5·0	456·4 ± 4·3	455·5 ± 3·0	457·6 ± 4·9
2nd cut	449·3 ± 6·2	461·0 ± 4·4	444·0 ± 6·8	445·5 ± 11·1	450·9 ± 9·7
Hydrogen (g kg^{-1} DM)					
1st cut	55·44 ± 0·53	56·13 ± 0·83	54·85 ± 0·69	54·50 ± 1·07	55·25 ± 0·58
2nd cut	55·78 ± 0·83	56·00 ± 0·76	52·86 ± 0·69	52·63 ± 1·41	54·06 ± 1·73
Nitrogen (g kg^{-1} DM)					
1st cut	11·4 ± 1·1	25·9 ± 4·5	21·6 ± 2·1	18·5 ± 2·9	16·6 ± 3·6
2nd cut	16·4 ± 2·1	28·8 ± 4·2	17·4 ± 3·5	17·8 ± 1·8	18·4 ± 3·6
Lignin (g kg^{-1} DM)					
1st cut	49·2 ± 11·7	74·9 ± 12·7	80·3 ± 15·2	72·1 ± 11·9	66·6 ± 10·7
2nd cut	45·1 ± 8·1	86·9 ± 18·1	101·2 ± 13·8	82·2 ± 11·5	80·7 ± 15·4
Ash (g kg^{-1} DM)					
1st cut	68·0 ± 4·5	73·0 ± 10·7	86·6 ± 8·6	86·1 ± 6·4	76·9 ± 9·0
2nd cut	103·6 ± 10·1	91·3 ± 11·3	120·5 ± 13·2	124·3 ± 20·7	105·0 ± 19·3

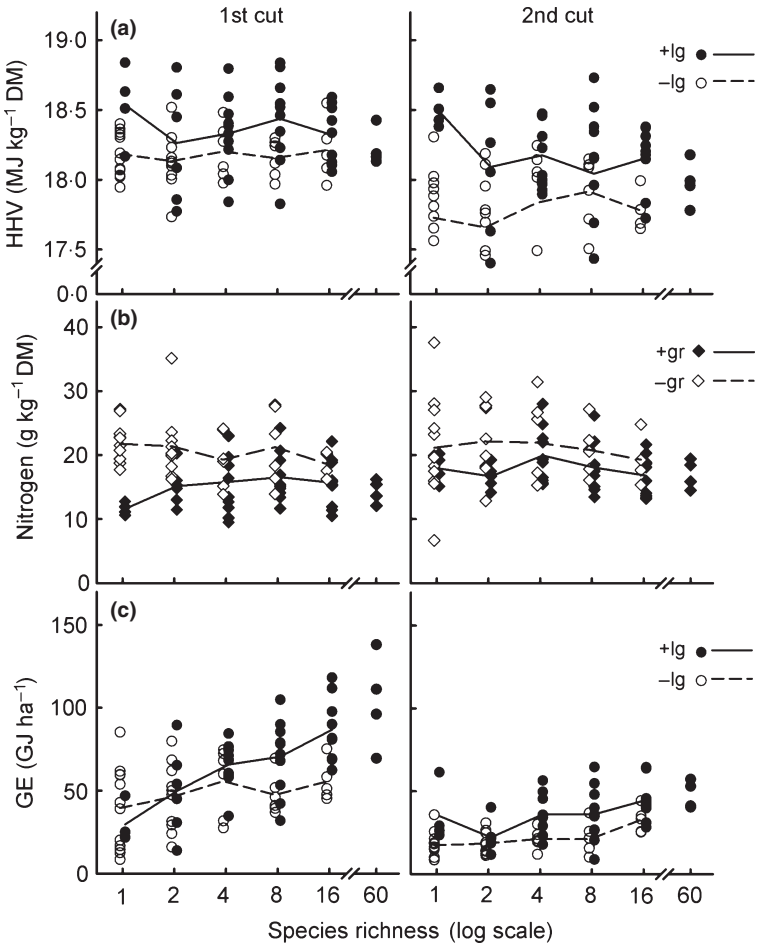

Figure 2 (a) Higher heating value (HHV), (b) nitrogen content and (c) gross energy yield (GE) in the 1st (late May) and 2nd cut (late August) at different levels of species richness. Mean values of species-richness levels are indicated by a solid line for plots with legumes present (+lg) or grasses present (+gr); mean values of plots with legumes absent (−lg) or grasses absent (−gr) are indicated by a dashed line. Values are means of both years.

Figure 2). Other functional-group effects on HHV were less consistent, i.e., the presence of grasses in the first cut and the presence of small herbs in the second cut both decreased HHV (Table 2).

Across all mixtures, C content ranged from 396 to 477 g kg^{-1} DM with an overall mean of 454 g kg^{-1} DM. In the functional-group monocultures, legumes had the highest C values in both cuts while all other functional groups were equally low (Table 1). C was affected by diversity parameters in the same way as HHV (Table 2), and values were slightly higher in the first than in the second cut (Table 1). Analysing the two cuts in separate models, the increase of C through the presence of legumes was most pronounced in both cuts (Table 2).

The content of H ranged from 46 to 58 g kg^{-1} DM, with an overall mean of 55 g kg^{-1} DM. In the functional-group monocultures, both legumes and grasses were slightly higher than tall and small herbs (Table 1). The only diversity parameter affecting H in the repeated measure model was FGC (Table 2A). Cut was affecting H, resulting in higher H values in the first than in the second cut (Table 1). Regarding functional-group effects, the presence of tall and small herbs had a decreasing effect on H content in both cuts, while grasses had an increasing effect in the second cut (Table 2).

The N content ranged from 8·9 to 38·4 g kg^{-1} DM with an overall mean of 18·7 g kg^{-1} DM. Highest N values among the functional-group monocultures were found in the legume plots, and lowest values occurred in the grass plots (Table 1). N was affected by SR and FGC in the repeated measures model (Table 2A). Being affected by the cut, N values were lower during the early growth period than during the late one (Table 1). N content was also affected by SR in both cuts, with a decline from 19·4 to 16·4 to 14·3 g kg^{-1} DM (from SR = 1 to 16 to 60) in the first cut, and in the second cut from 21·3 to 17·5 to 16·8 g kg^{-1} DM, respectively. In both cuts, the presence of legumes resulted in an increase of N, whereas the presence of grasses reduced it (Figure 2, Table 2).

Table 2 Analysis of variance of higher heating value (HHV) (MJ kg^{-1} DM), carbon, hydrogen, nitrogen and lignin content (all in g kg^{-1} DM) as functions of diversity parameters (species richness (SR), functional-group richness (FGR), functional-group composition (FGC) and with presence/absence of individual functional groups. All values are means of both years. (A) The overall effects of diversity and the significance of cut were tested. The effects of individual functional groups were tested separately for (B) 1st cut and (C) 2nd cut. Arrows indicate an increase (↑) or a decrease (↓) with the presence of the respective functional group.

		HHV		Carbon		Hydrogen		Nitrogen		Lignin	
Factor	**DF**	***F***	***P***	***F***	***P***	***F***	***P***	***F***	***P***	***F***	***P***
(A)											
Block	3	4·2	0·010	4	0·010	8	<0·001	<1	0·781	7	0·001
Log (SR)	1	0·9	0·340	3	0·083	<1	0·957	14	<0·001↓	<1	0·879
FGR	1	0·2	0·633	<1	0·710	<1	0·617	<1	0·679	<1	0·853
FGC	11	5·0	<0·001	3	0·002	6	<0·001	15	<0·001	9	<0·001
Residuals	59										
Repeated measures											
Cut	1	56	<0·001	48	<0·001	35	<0·001	11	0·002	42	<0·001
Cut × Log (SR)	1	<1	0·877	<1	0·877	<1	0·613	<1	0·535	1	0·293
Cut × FGR	1	2	0·212	1	0·383	<1	0·806	3	0·109	4	0·066
Cut × FGC	11	2	0·154	1	0·315	2	0·016	3	0·003	2	0·094
Residuals	62										
(B)											
Block	3	6	0·002	5	0·002	8	<0·001	2	0·073	4	0·012
Log (SR)	1	1	0·243	4	0·047	<1	0·611	5	0·037↓	<1	0·483
Legumes	1	16	<0·001↑	12	0·001↑	4	0·052	20	<0·001↑	15	<0·001↑
Grasses	1	13	0·001↓	7	0·010↓	2	0·147	36	<0·001↓	39	<0·001↓
Tall herbs	1	3	0·079	3	0·113	9	0·003↓	1	0·277	<1	0·828
Small herbs	1	2	0·186	3	0·110	5	0·023↓	1	0·366	4	0·057
FGC	8	2	0·123	1	0·267	1	0·397	5	<0·001	1	0·664
Residuals	59										
(C)											
Block	3	2	0·152	2	0·095	4	0·014	2	0·108	6	0·001
Log (SR)	1	<1	0·595	1	0·248	<1	0·841	13	0·001↓	<1	0·728
Legumes	1	22	<0·001↑	12	0·001↑	3	0·099	92	<0·001↑	17	<0·001↑
Grasses	1	<1	0·489	<1	0·747	7	0·011↑	9	0·005↓	31	<0·001↑
Tall herbs	1	2	0·217	1	0·309	12	0·001↓	2	0·213	<1	0·983
Small herbs	1	6	0·021↓	4	0·051	10	0·003↓	11	0·001↓	24	<0·001↓
FGC	8	1	0·341	1	0·437	2	0·022	2	0·033	3	0·009
Residuals	59										

Lignin ranged from 14 to 136 g kg^{-1} DM across both years with an overall mean of 74 g kg^{-1} DM. In the functional-group monocultures, lignin was highest in the small-herb plots followed by legumes, tall herbs and grasses (Table 1). Lignin was affected by cut (Table 2), and lignin contents were lower in the first than in the second cut (Table 1). Lignin content was not affected by SR but increased because of the presence of legumes in both cuts. By contrast, the presences of grasses had a decreasing effect on lignin in the first cut and an increasing effect in the second cut. Small herbs increased lignin content in the second cut but had no effect in the first cut (Table 2).

Biomass and GE yield

Gross energy yield was calculated as the product of HHV and biomass yield and was therefore affected by their respective patterns. The correlation of GE and biomass yield was at R^2 = 0·99 in both cuts, highlighting the importance of biomass yield for GE. Across the species mixtures, biomass yield ranged from 0·2 to 8·1 t DM ha^{-1}, with an overall mean of 2·4 t DM ha^{-1}, and the GE of the single cuts, with an overall mean of 43 GJ ha^{-1}, ranged from 3 to 150 GJ ha^{-1}. Biomass yield was almost twice as high in the first cut (3·1 t DM ha$^{-1)}$ as in the second cut

Table 3 Parameter estimates for the multiple regression analysis of gross energy yield. Abundances for legumes, grasses, tall and small herbs are given as DM yield percentage.

Factor	Estimate	s.e.	*t* value	*P*
1st cut				
INTERCEPT	61·03887	5·3854	11·334	<0·001
Legumes (lg)	1·27363	0·2869	4·439	<0·001
lg^2	−0·01375	0·0030	−4·558	<0·001
Tall herbs (th)	−0·13372	0·0878	−1·523	0·132
Small herbs (sh)	−0·27605	0·0901	−3·064	0·003
th × sh	0·01874	0·0064	2·913	0·005
2nd cut				
INTERCEPT	25·7760	3·4485	7·475	<0·001
lg	0·83347	0·1698	4·909	<0·001
lg^2	−0·00727	0·0018	−4·104	<0·001
th	−0·06183	0·0514	−1·202	0·233
sh	−0·06966	0·0499	−1·395	0·167
th × sh	0·00985	0·0033	2·969	0·004

(1·6 t DM ha^{-1}) as was GE (57 GJ ha^{-1} and 29 GJ ha^{-1} respectively). In the separate analysis of the two cuts, GE and biomass were significantly affected by SR (all $P < 0·001$). However, the magnitude of increase in GE with increasing SR in the first cut was from 40 GJ ha^{-1} (SR = 1) to 104 GJ ha^{-1} (SR = 60), while it was only from 26 to 48 GJ ha^{-1} in the second cut (Figure 2). Legumes were the only functional group whose presence had an effect on GE and biomass in both cuts (1st cut: both $P = 0·004$; 2nd cut: both $P < 0·001$). In terms of functional-group abundance, legumes again were the only functional group with a noticeable effect (Table 3). According to the models established through multiple regression analysis for predicting GE ($R^2 = 0·33$ in the first and $R^2 = 0·38$ in the second cut), GE increased with increasing legume abundance up to 47% of DM in the first cut and up to 58% DM in the second (Figure 3a).

To better understand the underlying diversity mechanisms, possible effects of SR on functional-group abundance were investigated and the dominance structure was quantified. As intended by the experimental design, legume abundance was not affected by SR (1st cut: $P = 0·14$; 2nd cut: $P = 0·22$; Figure 3b), and none of the interactions between abundance of a functional group and SR showed any effect. The same was true for the abundance of grasses ($P = 0·46$ and $P = 0·94$, respectively), small herbs ($P = 0·86$ and $P = 0·54$) and tall herbs ($P = 0·51$ and $P = 0·46$). The Shannon evenness (E) as an indicator for the dominance structure of mixtures increased with increasing SR in both cuts (with $E = 0·54$; 0·48, 0·60, 0·64 and 0·67 in the first cut and $E = 0·53$, 0·58, 0·63 and 0·67 in the second cut, for SR = 2 (only in the first cut), 4, 8, 16 and 60, respectively).

The total annual GE, the sum of GE from both cuts, varied from 19 to 265 GJ ha^{-1} $year^{-1}$, with a mean of 86 GJ ha^{-1} $year^{-1}$. Total annual GE was affected by SR and increased consistently with increasing SR (59 ± 27, 116 ± 33, 152 ± 36 GJ ha^{-1} $year^{-1}$ at SR = 1, 16, 60, respectively; Figure 4). The presence of legumes increased total annual GE at all levels of SR (Figure 4). However, the increase of total annual GE owing to legumes was more pronounced at higher diversity levels (43 GJ ha^{-1} $year^{-1}$ at SR = 16) as compared to only 12 GJ ha^{-1} $year^{-1}$ at SR = 1.

Ash and ash-forming elements

The ash content among the species mixtures varied between 53 and 220 g kg^{-1} DM, with an overall mean of 94 g kg^{-1} DM. In the functional-group monocultures, both legumes and grasses had the lowest mean values, while contents were higher in tall and small herbs (Table 1). Ash was negatively affected by SR and affected by FGC in the repeated measures model (Table 4A). Cut had an effect on ash (Table 4A), and there was lower ash content in the first cut than in the second (Table 1). Analysing ash in two separate models, there were no effects of SR (Table 4) while FGC had an effect in the first cut. Most functional-group effects were not consistent between the cuts: ash content was lower with grasses in the first cut and with legumes in the second, and small herbs increased ash content in the first cut. The only consistent functional-group effect was tall herbs, increasing ash in both cuts (Table 4).

Across all mixtures, K ranged from 7·6 to 34·1 g kg^{-1} DM, with a mean of 17·7 g kg^{-1} DM. The K content was affected by the presence of legumes, which reduced the K concentrations in both cuts, whereas small herbs increased it (Table 4). K in the ash was lowest in legume monocultures, with 183 g kg^{-1} ash in the first cut and 120 g kg^{-1} ash in the second cut (Figure 5). Mean K concentrations in the ash were equally high in the other three functional-group monocultures and the all-functional-group mixtures, with 242–255 g kg^{-1} ash in the first cut and 149–168 g kg^{-1} ash in the second cut (Figure 5).

Ca concentrations in the biomass ranged from 2·9 to 37·1 g kg^{-1} DM, with a mean of 15·2 g kg^{-1} DM and were lower when grasses were present (Table 4). There was a strong correlation between Ca and the abundance of grasses (first cut: $r = −0·73$; second cut: $r = −0·81$). The presence of legumes increased Ca concentrations, irrespective of the cut (Table 4). Mean Ca concentrations in the ash were lowest in grass monocultures (first cut = 64 g kg^{-1} ash; second

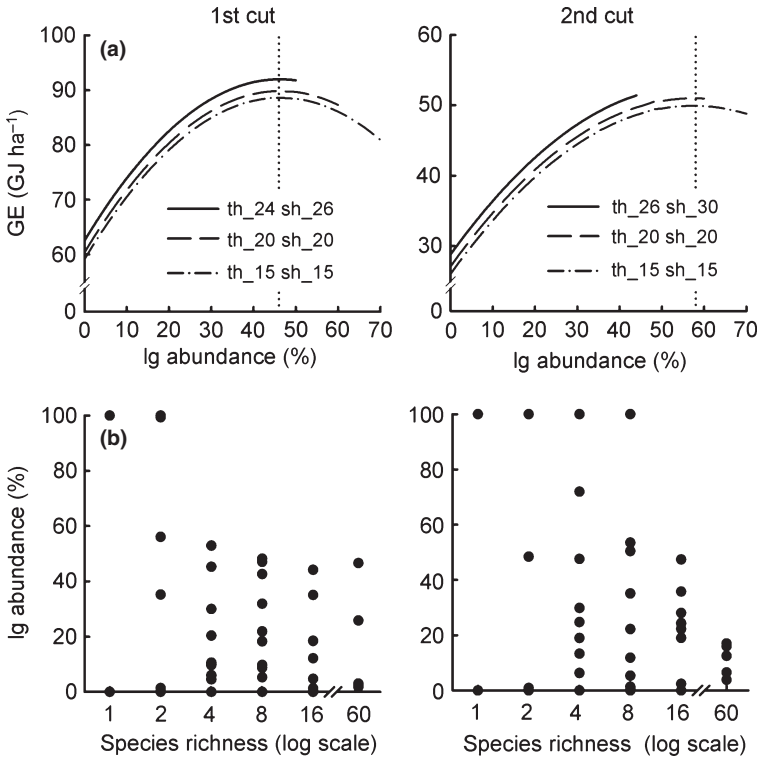

Figure 3 (a) Predictions of gross energy yield (GE) in the 1st (late May) and 2nd cut (late August) according to legume (lg) abundance (Table 3). The abundances of the other functional groups relevant for the multiple regression model [small herbs (sh) and tall herbs (th)) were varied (solid line indicates sh and th at their mean abundance; numbers indicate abundance in%]. Dotted lines indicate the point of highest GE according to the model. (b) Legume abundance in the biomass at different levels of species richness.

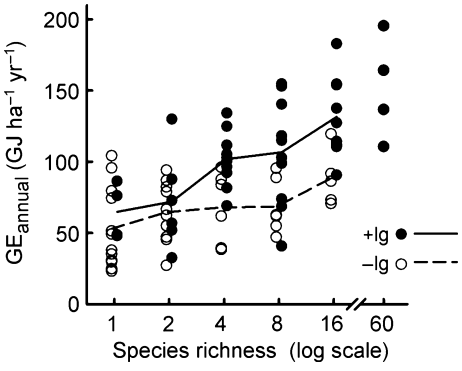

Figure 4 Annual gross energy yield (GE_{annual}) at different levels of species richness. Mean values of species-richness levels are indicated by solid line for plots with legumes present (+lg) and by dashed line for means with legumes absent (−lg). Values are means of both years.

cut = 110 g kg^{-1} ash) and highest in legume monocultures (first cut = 209 g kg^{-1} ash; second cut = 229 g kg^{-1} ash) (Figure 5). Mean Ca concentrations in the ash of all-functional-group mixtures were 140 g kg^{-1} ash in the first cut and 193 g kg^{-1} ash in the second (Figure 5).

Mg concentrations ranged from 0·8 to 7·2 g kg^{-1} DM, with a mean of 2·7 g kg^{-1} DM and, similar to Ca, were lower when grasses were present (Table 4). The correlation between Mg concentration and the abundance of grasses was $r = -0{\cdot}81$ in the first cut and $r = -0{\cdot}59$ in the second. The presence of legumes had a positive effect on the concentration of Mg, irrespective of the cut (Table 4). Mg was the only major ash-forming element that was affected by SR (Table 4, Figure 6). With increasing SR from 1 to 60, the mean values of Mg concentration in the biomass decreased by 30% in the first cut and by 22% in the second cut (Figure 6). Mean Mg concentrations in the ash were lowest in grass monocultures (first cut = 16·0 g kg^{-1} ash; second cut = 20·7 g kg^{-1} ash) and highest in legume monocultures (first cut = 47·5 g kg^{-1} ash; second cut = 45·9 g kg^{-1} ash) (Figure 5). Mean Mg concentrations in the ash of all functional-group mixtures were at 26·5 g kg^{-1} ash in the first cut and 30·1 g kg^{-1} ash in the second (Figure 5).

Cl and S concentrations

The Cl concentrations ranged from 0·9 to 9·2 g kg^{-1} DM, with a mean of 3·0 g kg^{-1} DM. The only functional-group effects on Cl concentration were weak and indicated a decline because of tall herbs, as well as an increase because of small herbs in the second cut (Table 4). Mean Cl concentrations did not vary much between the functional-group monocultures and were

Table 4 Analysis of variance of ash, potassium, calcium, magnesium, sulphur and chlorine content (all in g kg^{-1} DM) as functions of diversity parameters (species richness (SR), functional-group richness (FGR), functional-group composition (FGC) and with presence/absence of individual functional groups. All values are means of both years. (A) The overall effects of diversity and the significance of cut were tested. The effects of individual functional groups were tested separately for (B) 1st cut and (C) 2nd cut. Arrows indicate an increase (↑) or a decrease (↓) with the presence of the respective functional group.

		Ash		Potassium		Calcium		Magnesium		Sulphur		Chlorine	
Factor	**DF**	***F***	***P***	***F***	***P***	***F***	***P***	***F***	***P***	***F***	***P***	***F***	***P***
(A)													
Block	3	6	0·001	10	<0·001	7	<0·001	5	0·004	2	0·085	1	0·507
Log (SR)	1	6	0·020↓	1	0·261	<1	0·773	12	0·001↓	42	<0·001↓	<1	0·650
FGR	1	<1	0·872	<1	0·909	<1	0·503	<1	0·935	3	0·081	<1	0·690
FGC	11	4	<0·001	5	<0·001	13	<0·001	10	<0·001	3	0·011	2	0·129
Residuals	59												
Repeated measures													
Cut	1	236	<0·001	80	<0·001	462	<0·001	150	<0·001	409	<0·001	35	<0·001
Cut × Log (SR)	1	1	0·404	1	0·404	4	0·060	1	0·307	2	0·148	<1	0·710
Cut × FGR	1	<1	0·688	<1	0·644	2	0·140	1	0·350	2	0·221	<1	0·598
Cut × FGC	11	1	0·239	1	0·177	2	0·084	1	0·346	4	<0·001	1	0·370
Residuals	62												
(B)													
Block	3	3	0·022	8	<0·001	7	<0·001	4	0·013	7	<0·001	1	0·600
Log (SR)	1	4	0·059	3	0·106	1	0·409	10	0·003↓	35	<0·001↓	1	0·341
Legumes	1	<1	0·997	8	0·007↓	30	<0·001↑	68	<0·001↑	3	0·105	<1	0·578
Grasses	1	15	<0·001↓	6	0·016↓	69	<0·001↓	74	<0·001↓	21	<0·001↓	2	0·164
Tall herbs	1	9	0·004↑	3	0·092	6	0·015↑	<1	0·498	<1	0·817	2	0·194
Small herbs	1	16	<0·001↑	20	<0·001↑	4	0·041↑	<1	0·919	7	0·009↓	4	0·062
FGC	8	2	0·024	4	<0·001	3	0·006	5	<0·001	4	0·001	1	0·416
Residuals	59												
(C)													
Block	3	5	0·005	8	<0·001	7	0·001	5	0·003	2	0·146	1	0·250
Log (SR)	1	4	0·062	<1	0·587	1	0·273	8	0·006↓	23	<0·001↓	<1	0·896
Legumes	1	7	0·013↓	9	0·005↓	23	<0·001↑	18	<0·001↑	19	<0·001↓	<1	0·828
Grasses	1	1	0·324	2	0·205	31	<0·001↓	13	0·001↓	<1	0·741	<1	0·581
Tall herbs	1	6	0·020↑	2	0·157	22	<0·001↑	1	0·416	<1	0·998	5	0·032↓
Small herbs	1	3	0·092	16	<0·001↑	1	0·317	<1	0·578	<1	0·691	5	0·031↑
FGC	8	1	0·331	1	0·343	2	0·062	2	0·037	2	0·176	1	0·562
Residuals	59												

between 2·3 and 2·7 g kg^{-1} DM in the first cut and 2·6–3·3 g kg^{-1} DM in the second (Figure 7). The all-functional-group mixtures were in the same range.

The S concentrations ranged from 1·1 to 4·9 g kg^{-1} DM with a mean of 2·3 g kg^{-1} DM. S concentrations were significantly affected by SR (Table 4) and declined from 1- to 60-species plots by 23% in the first cut and by 16% in the second cut (Figure 6). Comparing the two cuts regarding functional-group presence, no consistent effects could be found. In the first cut, the presence of grasses and the presence of small herbs both increased S, while in the second cut the presence of legumes decreased it (Table 4). Therefore, S concentrations were lowest in the grass monocultures, with 1·5 g kg^{-1} DM in the first cut, and in the legume monocultures, with 2·6 g kg^{-1} DM in the second cut. The all-functional-group mixtures were as low as the lowest functional-group monocultures in the respective cut (Figure 7).

Discussion

Patterns of energy content and chemical composition

The energy content in the biomass of the Jena grassland communities measured as HHV was well within the range of values reported in other grassland studies

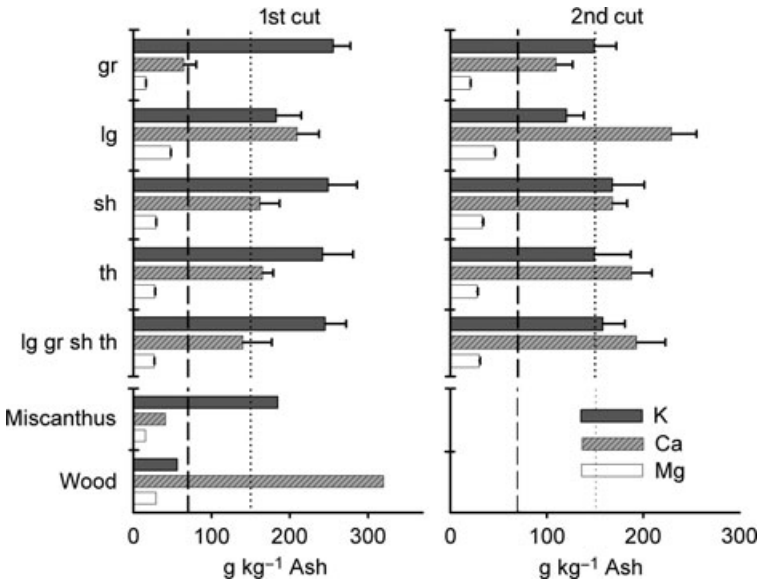

Figure 5 Concentrations of ash constituents in the ash of functional-group monocultures (gr = grasses, lg = legumes, sh = small herbs, th = tall herbs) and the all-functional-group mixtures, displayed separately for both cuts. Values are means of both years. Standard deviation is indicated by error bars. Ca, K and Mg are minerals relevant for ash-melting behaviour whereby the concentrations of Ca in the ash should be above 150 g kg^{-1} ash (dotted line) and K concentrations should be lower than 70 g kg^{-1} ash (dashed line), according to Obernberger *et al.* (2006). Values for wood (spruce as wood chips from Obernberger *et al.*, 2006) and miscanthus (calculated from Hartmann, 2009) are included for comparison.

(Table 5). However, in contrast to the approach of the Jena experiment, where emphasis was given to proper statistical methods leading to sound conclusions on the relationships between diversity parameters and energy content, their focus has been on investigating energy content along a spatial gradient across multiple sites. The results presented here clearly suggest that changes in SR do not affect the energy content of extensive grassland, as represented by the investigated assembled species communities investigated here, while the presence of legumes has a significantly positive effect on HHV (Figure 2a). A possible explanation for legumes having a positive effect on HHV may be their higher lignin content. Lignin is known to have a carbon content of 640 g kg^{-1} lignin, as compared to cellulose with a carbon content of 420 g kg^{-1} cellulose (Lewandowski, 2009), and therefore, a higher lignin content should result in a higher HHV. A study by Jung *et al.* (1997) observed similar patterns in lignin concentrations for legumes (77 g kg^{-1}DM) and grasses (33 g kg^{-1} DM) as found in this study. Looking at the relation between lignin content in the ash-free biomass and ash-free HHV, a positive correlation could be found in the first cut (R^2 = 0·34, P < 0·001), while it was weak in the second (R^2 = 0·08, P = 0·01). Better correlations might be found if lignin is entirely estimated through standard methods rather than by near-infrared spectroscopy. Andrés *et al.* (2005) discussed the difficulties with predicting ADL through near-infrared spectroscopy.

According to the formula that was used to calculate HHV, C is the most important element determining HHV, while N and H play a minor role in the equation. The range of C concentrations found in this study is lower than that found by Florine *et al.* (2006), which was between 484 and 537 g kg^{-1} DM. Of all the C concentrations found here, 75% were at or above 450 g kg^{-1} DM, and the overall mean of 454 g kg^{-1} DM is similar to data for grassland, which were reported by Hartmann (2009). The C content was negatively correlated to the ash content (R^2 = 0·65, P < 0·001).

Patterns and drivers of biomass and GE yield

Both biomass yield and GE showed relatively low minima compared with other studies. The lowest biomass presented here was 0·2 t DM ha^{-1}, which is largely because of the species monocultures and the functional-group monoculture plots. The lowest GE measured was 3 GJ ha^{-1}, which is about 30 GJ ha^{-1} lower than the lowest value observed by Tonn *et al.* (2010). Nevertheless, the experimental set-up used in this study allows a comparison with other diversity

Table 5 Mean, minimum (min) and maximum (max) values of energy content and combustion-relevant parameters from three different grassland experiments compared with values from the present study. Included are values from miscanthus and spruce wood, as examples of biomass frequently used for combustion.

		MJ kg^{-1} DM	g kg^{-1} DM						
Source and experimental set–up		**HHV**	**N**	**Ash**	**K**	**Ca**	**Mg**	**Cl**	**S**
Jena Experiment (2008–2009)	Mean	18·1	18·7	93	17·7	15·2	2·7	3·0	2·3
Extensive grassland, east	Min	16·3	8·9	53	7·6	2·9	0·8	0·9	1·1
Germany, one site, two harvests	Max	19·2	38·4	220	34·1	37·1	7·2	9·2	4·9
Tonn *et al.* (2010)	Mean	–	13·4	84	15·0	10·8	2·3	4·1	2·2
Extensive grassland, south–west	Min	18·2	10·3	61	8·4	6·7	1·1	1·7	1·5
Germany, six sites, different harvest dates	Max	19·1	17·4	99	24·2	14·9	3·7	11·2	3·5
Florine *et al.* (2006)	Mean	18·5	12·0	79	–	–	–	2·7	1·6
Perennial pastures, south Iowa,	Min	17·7	7·0	59	–	–	–	0·8	0·7
USA, ten sites, one harvest	Max	19·5	22·0	118	–	–	–	7·6	3·4
Richter *et al.* (2010)	Mean	17·8	13·7	–	12·0	10·0	3·2	4·8	1·6
Extensive grassland, Germany	Min	–	–	–	–	–	–	–	–
Five sites, one harvest	Max	–	–	–	–	–	–	–	–
Hartmann (2009)									
Miscanthus	Mean	19·1	7·3	39	7·2	1·6	0·6	2·2	1·5
Wood (spruce)	Mean	20·2	1·3	6	1·3	7·0	0·8	0·1	0·2

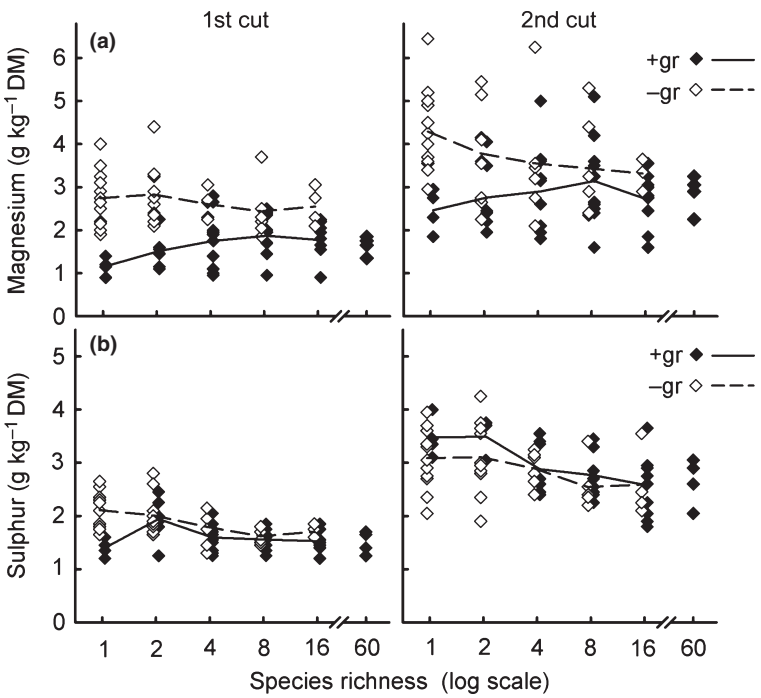

Figure 6 (a) Magnesium and (b) sulphur in the 1st (late May) and 2nd cut (late August) at different levels of species richness. Mean values of species-richness levels are indicated by solid line for plots with grasses present (+gr) and indicated by dashed line for plots with grasses absent (−gr). Values are means of both years.

studies such as the study by Tilman *et al.* (2006), representing a perennial grassland (with one removal of aboveground biomass per year through fire), where a range of 20 GJ ha^{-1} year^{-1} at SR = 1 up to 68 GJ ha^{-1} year^{-1} at SR = 16 was found. This is comparable to the GE values measured here in the first cut. However, GE of the monocultures had a mean of 40 GJ ha^{-1} year^{-1}, which is twice as high as those

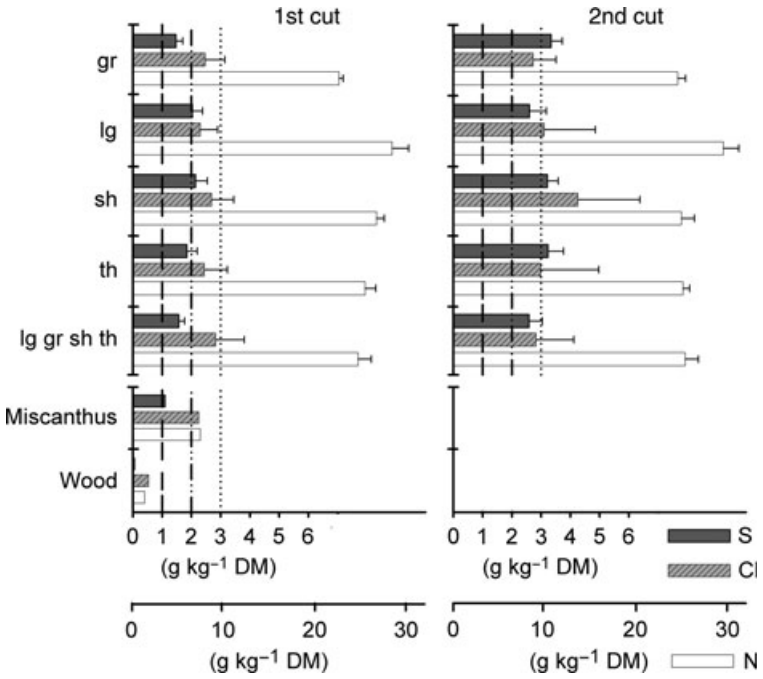

Figure 7 Concentrations of S, Cl and N in the biomass of the functional-group monocultures (gr = grasses, lg = legumes, sh = small herbs, th = tall herbs) and the all-functional-group mixtures, displayed separately for both cuts. Values are means of both years. Standard deviation is indicated by error bars. The concentrations of Cl should be below 1 g kg^{-1} DM (dashed line) to prevent corrosion and HCl emissions and below 3 g kg^{-1} DM (dotted line) to prevent emissions of polychlorinated dibenzodioxines and dibenzofuranes (PCDD/F). S concentrations should be below 1 g kg^{-1} DM (dashed line) to prevent corrosion and below 2 g kg^{-1} DM (dashed dotted line) to prevent SO_x emissions. N concentrations should be lower than 6 g kg^{-1} DM to prevent emission and corrosion (all thresholds according to van Loo and Koppejan, 2008). Comparison values included for wood (spruce, as wood chips) and miscanthus (van Loo and Koppejan, 2008).

observed by Tilman. This is largely because of the variation in biomass from the plant species grown in monoculture, because the assumed HHV of 18·5 MJ kg^{-1} by Tilman is almost identical to the mean HHV found in our study. The increase in annual GE from a one-species plot to a high-diversity 16-species plot was, on average, 108%, and even 172% from a one-species to a 60-species plot, while Tilman *et al.* (2006) reported increases of 157% and 238% respectively.

The experimental approach chosen here makes it possible to investigate further the underlying mechanisms driving the diversity–productivity relation, such as (i) 'sampling effect', describing the greater chance of including more productive species with increasing SR, (ii) 'facilitation', the presence of a species increases the growth or survival of another species or (iii) 'niche separation', the plant community can utilize more of the total nutrient pool and habitat space. The last two effects are often referred to as complementarity, so the question is whether diversity effects are driven by sampling or by complementarity.

If evenness (E; here the Shannon evenness index) declines (from $E = 1$ toward $E = 0$), a plant community is increasingly dominated by one high-yielding species. In this study, evenness increased slightly with SR, which indicates that high-diversity plots had a more balanced dominance structure than low-species plots. This is a weak indication for increasing importance of complementarity between the species at high SR. A more conservative and sound test for complementarity is by proving transgressive overyielding, as, by definition, in the case of transgressive overyielding, the productivity of a mixture is higher than the productivity of the most productive monoculture of the species contained in the mixture. Marquard *et al.* (2009) analysed 5 years (2003–2007) of biomass data from the Jena Experiment and found strong evidence for transgressive overyielding in a replacement series analysis on functional-group level.

Their results also showed that the complementarity effect, according to the method of additive partitioning by Loreau and Hector (2001), gained importance for the overall diversity effect over the years, while the 'selection effect' (species with higher than average monoculture yields dominate the mixtures, Loreau and Hector, 2001) declined over time. A further indication for complementarity being the main mechanism ruling diversity effects is when no functional groups interact with SR. This was tested here by fitting the presence of all four functional groups in the model before SR, which did not eliminate the significance of SR (1st cut: $P < 0{\cdot}001$; 2nd cut: $P = 0{\cdot}008$). In other words, SR affects biomass, independent of the presence of any functional group (including the legumes), which confirms similar findings in a biodiversity study by Tilman *et al.* (2001).

Looking at the complementarity effect more closely, the legume effect on biomass and GE is of utmost importance. Legumes had a significant positive effect on biomass and, as they were fitted after SR in the model, this effect was independent of SR. The evenness was not affected by the presence of legumes, indicating that the legume presence did not enhance total biomass through high legume-species biomass. Marquard *et al.* (2009) did not find any significant effect of legumes on the 'selection effect', which could explain the increase in biomass by a disproportionate contribution of legume biomass to the total biomass. Thus, it can be concluded that legumes affected biomass through facilitating the growth of other plants or through promoting complementary resource use and doing so in a similar pattern on all levels of SR. The facilitation mechanism of legumes is related to their ability to fix atmospheric N via symbiotic root bacteria. N_2-fixing plants often accumulate high concentrations of N in their tissue, which is released into the soil during their decomposition (Tomm *et al.*, 1995). Non N-fixing plants (grasses and herbs) can directly benefit from these processes by taking up the plant-available N released into the soil (Haynes, 1980; Spehn *et al.*, 2002). As N was found to be primarily limiting biomass production in the Jena Experiment (Oelmann *et al.*, 2007), it can be assumed that legumes have provided plant-available N for the system which promoted productivity and thereby GE. The highest productivity was observed at legume abundance of 47–58%, which is similar to the findings by Nyfeler *et al.* (2011) from a grassland study with moderate levels of fertilizer application (50 and 150 kg N ha^{-1} $year^{-1}$). Marquard *et al.* (2009) found the highest productivity after dividing legume abundance into five classes (0%, 0–33%, 33–66%, 66–100% and 100%) at an abundance between 66 and 100% in four of the 5 years.

Patterns in ash, ash-forming elements and ash-softening temperature

Ash contents are detrimental for combustion of biomass as they correlate negatively with the energy content (in this study, the correlation was $R^2 = 0{\cdot}54$; for rice straw, correlation was found to be $R^2 = 0{\cdot}46$ in the study by Jenkins *et al.*, 1996) and thus reduce energy output. Mean values of ash content found here were slightly higher than those found in other grassland studies (Table 5). While the maximum values in the latter studies only went up to around 100 g kg^{-1} DM, maximum values presented here were as high as 220 g kg^{-1} DM, similar to the 231 g kg^{-1} ash reported for biomass from grass verge (Hartmann, 2009). These high ash contents may partly be explained by the characteristics of the low-diversity plots, as the data set of this experiment (plots: $n = 76$) contained many plots with only one functional group ($n = 34$) of which several plots were species monocultures ($n = 16$). The biomass yield of these plots was often low, especially in the second cut, and had a large percentage of uncovered soil. Heavy rainfall on bare ground may have caused the contamination of plants with soil particles and thereby increased the ash content substantially. The higher ash contents in the second cut thus led to lower HHVs than observed in the first cut (see Table 1 and Figure 2a).

Another explanation for differences in ash contents may be found in the nutrient-uptake mechanisms of the plants or their morphology. The fact that functional-group monocultures of herbaceous species were higher in ash content than grasses and legumes (Table 3) is similar to the results observed by Tonn *et al.* (2010). Such patterns may occur because dicotyledonous herbaceous plants, owing to a high cation-exchange capacity of their roots, which often have a higher uptake of insoluble nutrients than monocotyledonous species (Drake *et al.*, 1951; Asher and Ozanne, 1961). For grass species, silicon is of great importance (Marschner, 2011) and it was found to largely contribute to the ash contents of grass-like species (65% of the total ash in switch grass, Jenkins *et al.* 1998). The strong increase of ash content in the grass functional-group monocultures from the first to the second cut may therefore be related to differences in silicon concentrations.

Morphological traits of the plants have been found to affect ash content, as higher nutrient concentrations are expected in the leaves than in the stems (Pahkala and Pihala, 2000). Furthermore, plants with relatively small growth, such as small herbs, might be much more prone to contamination with soil

particles after heavy rainfall events. Highest ash contents can therefore be expected in biomasses from grassland vegetation dominated by dicotyledonous, small-growing plants with a high leaf to stem ratio. Furthermore, high ash contents are to be expected in grass-dominated vegetation at late stages of maturity when silicon concentrations are increased (Tyler, 1971).

Besides discussing ash content itself, ash composition needs to be investigated as certain components that are negative for the combustion process and harmful to parts of the combustion chamber are contained in the ash. Common problems with ash from grassland biomass are, for example, slagging owing to low melting points of the ashes and corrosion of metal parts, which can reduce the combustion plant's lifetime (Obernberger *et al.*, 2006). Of the major ash-forming elements of biofuels, concentrations of K, Ca and Mg in the DM were analysed as they are known to affect AST. K-rich salts in combination with phosphates and silica were found to be responsible for low AST (Steenari *et al.*, 2009), and the ratio between K, on the one hand, and Ca and Mg on the other hand, indicate whether low or high AST can be expected. Furthermore, Hartmann (2009) suggested that K, Ca and Mg concentrations can be used to calculate AST.

The positive effect of legume presence on Ca and Mg, and the negative correlation of grass abundance with Ca and Mg observed in this study, are in line with the results from Tonn *et al.* (2010) and can partly be explained by the difference in cation exchange capacity, as discussed for ash content. However, the low K concentrations in the legume functional-group monocultures cannot be explained by this.

Whether the decline in Mg concentration with increasing SR was a true diversity effect and not merely coincidental with other variables was tested by fitting each functional-group term before SR in a sequence of analyses (results not shown). In both cuts, fitting grasses before SR removed the significance (Figure 6) while fitting any of the other functional groups before SR had no effect on the significance of SR. This means that the negative diversity effect regarding Mg could not be separated from the effect of grasses being present. This corresponds with the observation of different slopes in Mg concentration with increasing SR for grasses being present (+gr) or absent (−gr) (Figure 6). It is likely that this phenomenon was linked to the characteristics of +gr and −gr: the difference in Mg concentrations was most pronounced at SR = 1 where mono- and dicotyledonous species were fully separated (ratio of plots with monocotyledonous plants to plots with dicotyledonous plants to plots with a mixture of both was 1:3:0, see Figure 1), whereas at SR > 1, +gr and −gr did not segregate mono- and dicotyledonous plants any longer. While −gr only included dicotyledonous plants, +gr now consisted of either pure monocotyledonous plants (functional-group monocultures with grasses) or mixtures of both mono- and dicotyledonous plants (the ratio for SR = 2 was 1:5:2). At higher levels of SR, the number of mixed plots further increased (plot ratio at SR = 4 and 8 is 1:6:6 and 1:4:6 at SR = 16). Thus, at higher SR, +gr was very much affected by the presence of dicotyledonous plants, which results in a higher Mg content than a pure grass plot would have.

Comparing the concentrations of K, Ca and Mg in the ash of the functional-group monocultures with the recommended thresholds by Obernberger *et al.* (2006), the following trends were observed in both cuts (Figure 3): (i) all-functional-group monocultures, except for grasses, have a Ca concentration above the recommended 150 g kg^{-1} ash, which is acceptable with regard to AST, (ii) highest Ca concentrations were found in the legume monocultures, (iii) K concentrations generally exceeded the recommended value of 70 g kg^{-1} ash, therefore contributing to a lower AST, with lowest concentrations in the legume monocultures. Regarding the four functional-group mixtures, the same patterns with sufficient Ca and excessive K concentrations were found. Ca concentrations in the monocultures were higher than in miscanthus but lower than in wood (Obernberger *et al.*, 2006). With regard to the K concentrations, it must be considered that the temperature of the combustion plays a major role in the accumulation of K in the ash. Knudsen *et al.* (2004) found little to no K release into the gas phase at temperatures below 700°C, while 50–80% were released at temperature above 1100°C, depending on the substrate. Thus, at temperature < 700°C, K would mainly contribute to ash softening whereas at temperatures > 1100°C, it could become a major emission and fouling issue.

The equation suggested by Hartmann (2009) to estimate AST originates from biomass ashes (woody and herbaceous materials, $n = 67$) and it yields relatively accurate results ($R^2 = 0{\cdot}60$; s.e. = 88°C; Hartmann *et al.*, 2000). In accordance with the above-discussed effects of legumes and grasses on concentrations of K, Ca and Mg, it can be assumed that the presence of legumes would increase AST while grasses would decrease it. However, this equation is rather simple regarding the number of variables included and it may thus underestimate the complexity of chemical interactions that lead to ash-softening behaviour, as shown in other studies (Bryers, 1996; Seggiani, 1999).

Patterns in N, Cl and S content and their impacts on corrosion and emission

N contained in the vegetation plays a major role during combustion as it is the source of undesired NO_x emissions. The measured range of N concentrations was wider, and the mean N concentrations were slightly higher, than in other studies on extensive grasslands (Table 5). The decline of N concentrations with increasing SR was not independent of the presence of grasses as revealed by fitting other functional-group terms prior to SR in the model (data not shown) and by the non-parallel slopes of N concentration with increasing SR for +gr and −gr (Figure 2). Again, the characteristics of +gr and −gr may be a possible explanation for this effect, as was discussed for Mg.

N concentrations were above the critical level of 6 g kg^{-1} DM as suggested by Obernberger *et al.* (2006) and were more than twice as high as those of miscanthus (7·3 g kg^{-1} DM) and out of the range of values for wood from spruce (1·3 g kg^{-1} DM; van Loo and Koppejan, 2008) (Figure 7).

The range of Cl concentrations measured here was comparable to values found in other studies (Table 5). According to van Loo and Koppejan (2008), a critical value for Cl is 1 g kg^{-1} DM if corrosion and HCl emissions are to be prevented, while the Cl concentrations have to be below 3 g kg^{-1} DM to prevent PCDD/F emissions. All functional-group monocultures and mixtures exceeded the 1 g kg^{-1} DM, and the values in the second cut even exceeded the 3 g kg^{-1} DM threshold (Figure 7).

Sulphur contained in the biomass forms gaseous SO_x and alkali, as well as earth-alkali sulphates. If a sufficient amount of alkali and earth alkali metals, especially Ca, are present, more than 50% of the S can be fixed in the ash (e.g. as K_2SO_4 or $CaSO_4$) (Obernberger *et al.*, 2006). The S not fixed in the ash will either contribute to corrosion by releasing and replacing the Cl in alkali chlorides or cause SO_x emission. The S concentrations measured here were similar to the findings of Tonn *et al.* (2010) (Table 5). Critical values for S in biomasses are 1 g kg^{-1} DM to prevent corrosion and 2 g kg^{-1} DM to prevent SO_x emission (van Loo and Koppejan, 2008). None of the functional-group monocultures and mixtures was below 1 g kg^{-1} DM, while biomass from the first cut (except for small-herb monocultures), and from all monocultures and mixtures of the second cut (Figure 7), exceeded 2 g kg^{-1} DM. Testing whether the SR effect on S was intermingled with any other functional group by changing the order of terms entering the model confirmed that S was affected by SR independent of any other variable.

Tackling fuel-related problems and the advantage of high diversity

For grassland biomass as a solid fuel, it has been suggested there should be an extensive grassland management with one late cut and no application of fertilizers (Prochnow *et al.*, 2009). Regarding the biomass investigated in this study, it clearly shows that mineral contents (e.g. K, S, Cl) from an early cut are less favourable for combustion than the contents from a late cut. However, there are several ways to deal with fuel-quality deficiencies of grassland biomasses: (i) the combustion plant can be designed in a way that the amount of problematic flue gases is reduced through a staged combustion, while dust precipitation units (most relevant flue gas-cleaning step for solid fuels) further reduce emissions (Obernberger *et al.*, 2006), (ii) the substrate can be pre-treated prior to combustion through hydro-thermal conditioning and mechanical dewatering of nutrients to reduce ash concentration (Richter *et al.*, 2010, 2011); (iii) delayed harvest can improve fuel quality [in the case of extensive grassland harvested in February (Tonn *et al.*, 2007) and in the case of reed canary grass harvested in March (Christian *et al.*, 2002 and Xiong *et al.*, 2008)] mainly because of natural leaching through precipitation and loss of mineral-rich matter such as leaf biomass.

By hydro-thermal conditioning prior to mechanical dewatering through a screw press, ash constituents of extensive grasslands were reduced by 74–84% for K, by 79–89% for Cl, by 39–57% for S and by 19–43% for N, depending on the substrate and the conditioning temperature applied (Richter *et al.*, 2010, 2011). Reducing mineral concentrations prior to combustion through delayed harvest will result in similar fuel-quality improvements, but may cause large biomass losses in the field of 30–70% owing to particle breakdown, depending on the substrate and climatic conditions (Christian *et al.*, 2002; Tonn *et al.*, 2007). The effect of precipitation intensity as part of natural leaching has been studied by analysing samples after natural rain events (for straw by Jenkins *et al.*, 1996) and after simulated rain (for permanent grassland by Tonn *et al.*, 2011).

Considering that ash-related problems can partly be addressed by pre-treatment of the biomass and by adjusting the combustion plant, the most important and positive effect of high-diversity over low-diversity extensive grassland is the increase in energy yield. Looking closely at the diversity steps relevant for practice, the annual GE in this study increased by only 4% from 4 to 8 species, while it showed an increase of 26% from 8 to 16 species and increased by 65% from 8 to 60 species (Figure 4). It is evident that an

increased energy yield of between 26 and 65% will have a direct impact on the profitability of using extensive grassland biomass for energy production.

Experimental approaches and their relevance for extensive grassland farming

Diversity effects in grassland communities have been widely studied in artificially assembled communities of plant species. However, there has been a debate about whether the results of such studies can be applied to natural communities (Huston, 1997; Loreau *et al.*, 2001; Díaz *et al.*, 2003; Pfisterer *et al.*, 2004). Especially when interpreted in the framework of agricultural management, the relevance of these studies has to be assessed.

The clear advantage of studies with artificial communities is that SR or FGR can be precisely maintained, and the variance in abiotic conditions and community structure between replicates is kept small to yield sound statistical results. Regarding grassland management, their results may apply directly to, for example, the practice of ley farming in which crop growing is alternated by grassland for several years to improve soil fertility. However, Hopkins and Holz (2006) argued that further experiments should be more geared towards agricultural practices because many of the diversity–productivity experiments were conducted without fertilizer and other agricultural inputs and were conducted on prepared sites.

While Weigelt *et al.* (2009) studied diversity–productivity interactions in artificially assembled communities along a management and a fertilization gradient, others have suggested that removal experiments are the most suitable for understanding the role of functional groups under natural conditions (Díaz *et al.*, 2003; McLaren and Turkington, 2010). However, Díaz *et al.* (2003) concluded from a comparison of removal experiments and experiments with artificially assembled communities that both these experimental approaches should be used to complement each other, rather than one approach being favoured over the other.

The results from the present study have relevance for extensive, non-fertilized grassland, but the presented effects should be further examined with other methodologies that will shed light from different angles on relationships that cannot be assessed through experiments with artificial communities.

Conclusions

Using artificial grassland communities to identify diversity effects on fuel quality was successful and supports the following conclusions: (i) SR does not affect the energy potential of extensive grassland by modifying the energy content directly; rather, it is the increase in biomass that makes high-diversity grassland more valuable for bioenergy production, (ii) legumes increase GE yield mainly by increasing biomass owing to facilitation and complementary resource use and, to a smaller extent, by increasing energy content owing to higher carbon contents, (iii) identifying one functional group as being solely beneficial or disadvantageous for fuel quality is impossible, as they exhibit contrasting effects with regard to different parameters, for example, legumes increase Ca concentrations (increasing AST) and, at the same time, increase N concentrations (resulting in higher NO_x emissions), whilst grasses have lower N concentrations (lower NO_x emissions) but the highest K concentrations (decreasing AST), (iv) several positive aspects of high SR regarding ash properties exist, such as lower N and S concentrations, even though they could not always be fully disentangled from functional-group effects.

It was further concluded that to facilitate a high energy output in extensive grassland systems, the defoliation and nutrient management should be focused on the maintenance of highly diverse, legume-based grassland communities. Given the high concentrations of corrosion and emission-related constituents, a treatment of the fuel through nutrient leaching as well as technical adaptation of the combustion plants is recommended for grassland biomass at all levels of diversity.

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